

Scale-Explicit and Uncertainty-Qualified Assessment of Tangent-Only Matrix Perturbations in Coupled Finite-Element Linearizations

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Abstract

Tangent-only matrix perturbations are often judged by their norm relative to a finite-element tangent, although the realized directional response also depends on inverse amplification, probe direction, coordinate representation, and finite-precision resolvability. Existing perturbation, scaling, and floating-point tools do not by themselves provide a coordinate-explicit procedure for qualifying the response of a selected coupled linearization. We develop a scale-explicit and uncertainty-qualified assessment that declares the solved space and coordinates, separates matrix, operator, and realized directional measures, and evaluates each arithmetic route against its own error enclosure. Its mathematical basis combines an exact finite response identity, consistent transformation of the operator and perturbation, and a zero-exclusion criterion under route-specific uncertainty. Pre-specified synthetic families verified a six-amplitude directional counterexample, coordinate transformation and redeclaration, and complete development/holdout uncertainty coverage under decimal-source and binary64-operand truth contracts. The procedure was then applied separately to two independently frozen coupled finite-element linearizations of dimensions 196 and 900. Direct-difference and response-equation routes excluded zero in both applications under their respective route-specific uncertainties. These results show that matrix-space smallness alone does not determine a realized response and that numerical distinguishability re-

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quires explicit coordinate and arithmetic-route declarations. The two applications demonstrate replication of the procedure, not guard safety, nonlinear prediction, or generality across finite-element systems.

Keywords: tangent matrix, matrix perturbation, finite-element linearization, coordinate scaling, floating-point uncertainty, directional response

1. Introduction

Coupled nonlinear finite-element analyses repeatedly form local or global tangent systems whose blocks represent variables with different units and characteristic scales. Implementations may add a small diagonal or structured matrix term to avoid singular factorization, regularize an intermediate solve, or continue an iteration through a difficult local linearization. We call such a change a *tangent-only matrix perturbation* when it modifies the matrix supplied to the linear solve without adding the corresponding term to the residual. This object is narrower than residual-consistent finite-element stabilization and should not inherit claims about consistency, mixed-method stability, conservation, or boundedness. Field-aware solvers and matrix normalization are active computational choices in coupled finite-element systems [1, 2, 3, 4]; the response induced by a tangent-only change therefore requires a separate assessment.

A matrix perturbation is commonly described as small when $\|\Delta J\|/\|J\|$ is small. That ratio answers how much the declared matrix changed in one norm and coordinate system, but not how one selected right-hand side responds. The realized change also depends on inverse propagation and on the direction of ΔJz . Equal matrix-space amplitudes can consequently produce different solution responses, while a condition number supplies a worst-case sensitivity comparison rather than a directional explanation [5, 6, 7]. This distinction matters in mixed-field mechanics because the same numeric diagonal coefficient can act beside displacement, pressure, or internal-variable blocks whose coordinates and residual scales differ by many orders of magnitude.

Matrix perturbation theory provides exact response identities and norm bounds; conditioning theory separates forward sensitivity from backward error; scaling and equilibration expose coordinate dependence; and floating-point analysis explains why subtracting two accurate parent solutions may not resolve their small difference [6, 7, 8, 9, 10, 11, 12, 13]. Verified computing and precision-aware refinement offer routes for constructing reference enclosures [14, 15, 16].

30 These bodies of theory are essential, but they do not by themselves form a
31 coordinate-explicit, probe-aware, route-specific procedure for qualifying the
32 numerical response induced by a tangent-only perturbation in a selected coupled
33 finite-element linearization.

34 The computational gap is therefore one of integration and declaration. An
35 interpretable assessment must fix the solved space, unknown and residual coordi-
36 nates, norm, right-hand side, perturbation operator, arithmetic response route, and
37 numerical truth object before comparing values. It must then keep matrix ampli-
38 tude, inverse-mediated amplification, and realized response distinct, and it must
39 qualify the observed response with an error enclosure matched to the operations
40 that produced it. Without these declarations, a small coefficient, a small backward
41 error, or a high-precision difference can each answer a different question while
42 appearing to support the same conclusion.

43 We propose a scale-explicit and uncertainty-qualified computational assess-
44 ment procedure for this purpose. The procedure consistently transforms J , ΔJ ,
45 and b ; evaluates matrix, operator, and directional quantities in declared coordi-
46 nates; applies the exact finite response relation; builds an uncertainty budget for
47 the selected arithmetic route; and classifies only whether the resulting response
48 interval excludes zero. Figure 1 gives the conceptual sequence, and Algorithm 1
49 supplies its executable presentation. The procedure does not select a perturbation
50 or define a universal acceptance threshold.

51 The contribution has four parts. First, an exact two-dimensional construc-
52 tion proves that matched matrix amplitude, condition number, and finite worst-
53 direction amplification do not determine a realized directional response. Second,
54 a coordinate analysis distinguishes transformation of one fixed perturbation from
55 redeclaration of a same-numeric identity after rescaling. Third, a two-contract un-
56 certainty model separates decimal-source and binary64-operand truth objects and
57 qualifies direct-difference and response-equation routes independently. Fourth,
58 the complete procedure is validated on pre-specified synthetic families and applied
59 independently to two frozen coupled finite-element tangent systems.

60 The validation hierarchy is deliberately layered. Analytical constructions ver-
61 ify the directional and coordinate statements. Development and disjoint holdout
62 families test uncertainty coverage within specified operator classes and dimen-
63 sions. The two finite-element applications then use immutable 196- and 900-
64 dimensional binary64 operand snapshots, independent high-precision references,
65 and route-specific error enclosures. These applications demonstrate replication on

two coupled linearizations; they are not a mesh sequence or a statistical sample of finite-element systems.

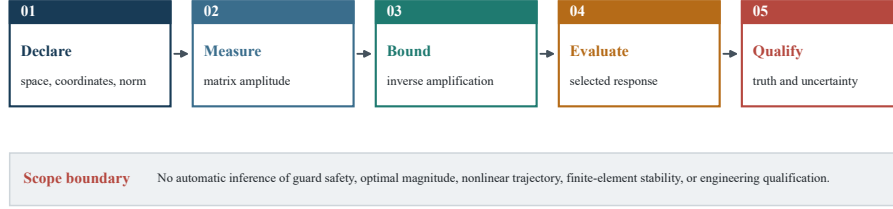


Figure 1: Computational sequence from solved-space and coordinate declaration to route-specific uncertainty-qualified response interpretation.

Section 2 defines the mathematical and computational framework. Sections 3–5 verify the directional construction, coordinate transformation, and uncertainty model. Section 6 applies the procedure to the two coupled finite-element linearizations, and Section 7 discusses interpretation, scalability, and limitations.

2. Mathematical and Computational Framework

2.1. Problem definition and solved-space contract

Consider an unguarded linear system and a matrix-perturbed system posed on the same solved space. The matrices J and $J + \Delta J$ are assumed square and nonsingular, both systems use the same right-hand side b , and their solutions are represented in one fixed coordinate system. A constrained problem may enter this framework only after its free or otherwise reduced solved space has been declared; no assertion is made for an unreduced singular operator. We write $\|\cdot\|$ for one declared vector norm and its associated subordinate matrix norm, so all matrix-vector and matrix-matrix product bounds use the same convention. The comparison therefore concerns two operators acting on the same data, not two problems with different loads, coordinates, nullspaces, or residual roots. These requirements are the minimum conditions under which a finite solution-response identity and its norm bounds have a single interpretation [6, 7].

2.2. Matrix, operator, and directional measures

The unguarded solution is denoted by z , the guarded solution by z_g , and their difference by $\delta z = z_g - z$. Six quantities are retained because they answer differ-

ent questions. The matrix ratio μ_J measures perturbation amplitude in the declared coordinates; ρ_1 measures first-order worst-direction amplification through the unguarded inverse; ρ_g measures finite guarded-inverse amplification; η_z measures the realized relative response for the selected right-hand side; G_g reports the diagnostic ratio between that realized response and matrix amplitude when μ_J is nonzero; and $\kappa(J)$ measures normwise sensitivity of the unguarded linear problem. None of these quantities substitutes for another. In particular, an operator norm is not the response of a selected probe, and a condition number is not a causal classification of an observed change [5, 7]. Taken together, the quantities define an assessment chain rather than a composite guard-quality score.

Figure 2 places a tangent-only perturbation within a generic mixed-field block operator without assigning physical meaning to one scalar diagonal coefficient.

Mixed-field tangent and tangent-only numerical perturbation

	u_x	u_y	p	S	
u_x	J_{u_x}	$J_{u_x u_y}$	$J_{u_x p}$	$J_{u_x S}$	$J_{\text{returned}} = J_{\text{phys}} + \Delta J_{\text{num}}$
u_y	$J_{u_y u_x}$	J_{u_y}	$J_{u_y p}$	$J_{u_y S}$	Red diagonal markers
p	$J_{p u_x}$	$J_{p u_y}$	J_p	$J_{p S}$	tangent-only perturbation overlay
S	$J_{S u_x}$	$J_{S u_y}$	$J_{S p}$	J_S	One scalar coefficient does not define one scale across mixed fields.

u_x : displacement [m] u_y : displacement [m] p : pressure [Pa] S : saturation [-]

Conceptual anatomy only; no UEL revalidation claim.

Figure 2: Generic mixed-field tangent with a tangent-only diagonal perturbation shown as a separate matrix overlay.

Table 1 lists the retained metrics and their non-substitution rules.

$$\begin{aligned}
 Jz &= b, & (J + \Delta J)z_g &= b, & \delta z &= z_g - z, \\
 \mu_J &= \frac{\|\Delta J\|}{\|J\|}, & \rho_1 &= \|J^{-1}\Delta J\|, & \rho_g &= \|(J + \Delta J)^{-1}\Delta J\|, \\
 \eta_z &= \frac{\|\delta z\|}{\|z\|}, & G_g &= \frac{\eta_z}{\mu_J} \quad (\mu_J > 0), & \kappa(J) &= \|J\| \|J^{-1}\|.
 \end{aligned} \tag{1}$$

Table 1: Retained metrics and their non-substitution rules.

Metric	Definition	Question answered	Does not establish
μ_J	$\ \Delta J\ /\ J\ $	Matrix-space amplitude	Realized response
ρ_1	$\ J^{-1}\Delta J\ $	First-order worst-direction amplification	Finite selected-probe response
ρ_g	$\ (J + \Delta J)^{-1}\Delta J\ $	Finite worst-direction amplification	Response for one right-hand side
η_z	$\ z_g - z\ /\ z\ $	Realized response for one probe	Worst-case behavior
G_g	η_z/μ_J	Diagnostic response/amplitude ratio	Condition number or cause
$\kappa(J)\mu_J$	$\ J\ \ J^{-1}\ \mu_J$	Worst-case normwise comparison	Directional mechanism or trajectory

Note: Every metric requires a declared solved space, coordinates, compatible norm, right-hand side, perturbation route, and truth contract as applicable. Full declarations are listed in the Supplementary Information.

103 2.3. Exact finite perturbation response

104 Proposition P1 follows by writing the perturbed solution as $z + \delta z$ and subtract-
 105 ing the unguarded equation from the perturbed equation. The common right-hand
 106 side cancels, leaving a finite relation between the matrix perturbation and the ex-
 107 act solution response. Nonsingularity of $J + \Delta J$ then gives the guarded-inverse
 108 form without a small-perturbation approximation. This identity is algebraic and
 109 does not require symmetry or positive definiteness; its scope is fixed instead by
 110 the common-system assumptions stated above. It also identifies the direction that
 111 drives the response: the perturbation first acts on the unguarded solution through
 112 $\Delta J z$, after which the guarded inverse propagates that vector. The identity therefore
 113 separates a declared matrix change from the response of a particular probe, while
 114 remaining silent about floating-point accuracy of a computed difference [6, 7].

$$115 \quad \begin{aligned} (J + \Delta J)\delta z &= -\Delta J z, \\ \delta z &= -(J + \Delta J)^{-1}\Delta J z. \end{aligned} \quad (2)$$

116 Taking the associated vector norm and subordinate matrix norm in the exact
 117 identity yields the finite guarded-inverse response bound. For nonzero z , the re-
 118 alized relative response satisfies $\eta_z \leq \rho_g$. The inequality is a worst-direction
 119 bound because ρ_g maximizes amplification over admissible directions in the se-
 120 lected norm, whereas η_z evaluates only the direction generated by the declared
 121 solution and perturbation. Equality may occur for an aligned direction, but it is
 122 not implied for an arbitrary right-hand side. This distinction is important even
 123 when ρ_g is evaluated accurately: the operator quantity answers how large the re-
 124 sponse could be over directions, not what the chosen probe did. The bound is
 125 finite whenever the perturbed operator is nonsingular, including cases for which
 126 the first-order Neumann condition below is unavailable.

$$127 \quad \eta_z \leq \rho_g. \quad (3)$$

128 For a comparison based on the unguarded inverse, define $E = J^{-1}\Delta J$. Then
 129 $J + \Delta J = J(I + E)$, and the finite response operator can be written as $(I + E)^{-1}E$.
 130 If the subordinate norm satisfies $\rho_1 = \|E\| < 1$, the Neumann series establishes
 131 nonsingularity of $I + E$ and bounds its inverse by $1/(1 - \rho_1)$. Consequently, the
 132 realized response is bounded by $\rho_1/(1 - \rho_1)$. The denominator is part of the finite
 133 theorem and cannot be dropped. The shorter comparison with ρ_1 alone is only a
 134 first-order interpretation as ρ_1 tends to zero, not a finite bound at arbitrary pertur-

135 bation size. If ρ_1 is at least one, the exact identity may still hold because both
 136 matrices can remain nonsingular, but this Neumann argument supplies no bound
 137 [6, 7].

$$138 \quad \rho_1 < 1, \quad \eta_z \leq \frac{\rho_1}{1 - \rho_1}. \quad (4)$$

139 The matrix-amplitude ratio enters a looser comparison through submultiplica-
 140 tivity. Under the same norm, ρ_1 is no greater than $\|J^{-1}\|\|\Delta J\|$, which equals $\kappa(J)\mu_J$.
 141 If the product is below one, substitution into the Neumann bound gives a finite
 142 condition-number comparison with the denominator retained. This result is norm-
 143 wise and worst case. It depends on the declared coordinates because both $\kappa(J)$ and
 144 μ_J change when unknown and residual scales change, and it does not encode the
 145 direction ΔJz . A small value can support a worst-case upper estimate under the
 146 assumptions; a large value can indicate that the estimate is uninformative. Neither
 147 outcome identifies the mechanism of a particular observed response. Condition
 148 theory therefore limits what can be inferred from matrix amplitude, rather than
 149 replacing the directional calculation [5, 7].

$$150 \quad \begin{aligned} \rho_1 &\leq \kappa(J)\mu_J, \\ \kappa(J)\mu_J < 1 &\implies \eta_z \leq \frac{\kappa(J)\mu_J}{1 - \kappa(J)\mu_J}. \end{aligned} \quad (5)$$

151 The framework has four explicit exclusions. First, changing b adds a data-
 152 perturbation term, so the fixed-right-hand-side identity no longer describes the
 153 full response. Second, changing the residual equation or its root is not a tangent-
 154 only comparison. Third, a singular unreduced operator requires a declared quo-
 155 tient space, nullspace projection, generalized inverse, or other well-posed reduced
 156 formulation before these metrics can be interpreted; saddle-point and constrained
 157 systems cannot be treated as ordinary nonsingular systems without that declaration
 158 [1, 2]. Fourth, the identities compare two local linear solves at one state. They do
 159 not determine subsequent Newton acceptance, iteration count, globalization path,
 160 branch sequence, basin, or final root, all of which require additional nonlinear as-
 161 sumptions and evidence [17, 18].

162 2.4. Computational assessment procedure

163 Algorithm 1 formalizes the existing assessment chain as a ten-stage computa-
 164 tional procedure. Coordinate transformation is developed and verified in Section 4,

165 while the route-specific uncertainty construction is developed and validated in Sec-
 166 tion 5. The algorithm introduces no metric or decision beyond Eqs. (1) and (17);
 167 its only classification test is whether the declared response interval excludes zero.
 168 Figure 1 remains the conceptual overview, whereas Algorithm 1 states the exe-
 169 cutable order of declarations and evaluations.

Algorithm 1 Scale-explicit uncertainty-qualified assessment of a tangent-only matrix perturbation.

Require: J , ΔJ , b ; solved-space declaration; D_x , D_r ; a compatible norm; truth contract; response route; precision and uncertainty specification

Ensure: Route-specific qualified response statement

- 1: Declare the solved space, coordinates, norm, right-hand side, and perturbation route.
 - 2: Transform the operator, perturbation, right-hand side, and response coordinates consistently using D_x and D_r .
 - 3: Evaluate the matrix-space amplitude μ_J in the declared coordinates.
 - 4: Evaluate available inverse-mediated quantities, including ρ_1 or ρ_g , without substituting them for a directional response.
 - 5: Evaluate the selected realized directional response d_{obs} by the declared arithmetic route.
 - 6: Declare the numerical truth contract for the operands and reference response.
 - 7: Construct the route-specific parent, arithmetic, and subtraction uncertainty components.
 - 8: Form the response interval $[\max(0, d_{\text{obs}} - U), d_{\text{obs}} + U]$.
 - 9: Test the sole distinguishability condition $d_{\text{obs}} - U > 0$.
 - 10: Report the matrix metric, operator metric, directional response, uncertainty qualification, and scope boundary.
-

170 3. Analytical Verification

171 Proposition P2 fixes the Euclidean vector norm and spectral matrix norm and
 172 considers a two-dimensional identity operator. The right-hand side is the first co-
 173 ordinate vector, so the unguarded solution is $z = e_1$. Two rank-one diagonal per-
 174 turbations are then formed with the same positive amplitude ε : case A acts on the
 175 first coordinate, and case B acts on the second coordinate. The construction holds
 176 the operator, norm, right-hand side, perturbation amplitude, and condition number

177 fixed while changing only the orientation of the perturbation relative to the se-
 178 lected solution. Because both perturbations are scaled orthogonal projectors, their
 179 spectral norms and worst-direction amplification norms can be calculated exactly.
 180 This analytical construction isolates directional dependence without invoking a
 181 statistical ensemble or a finite-element discretization.

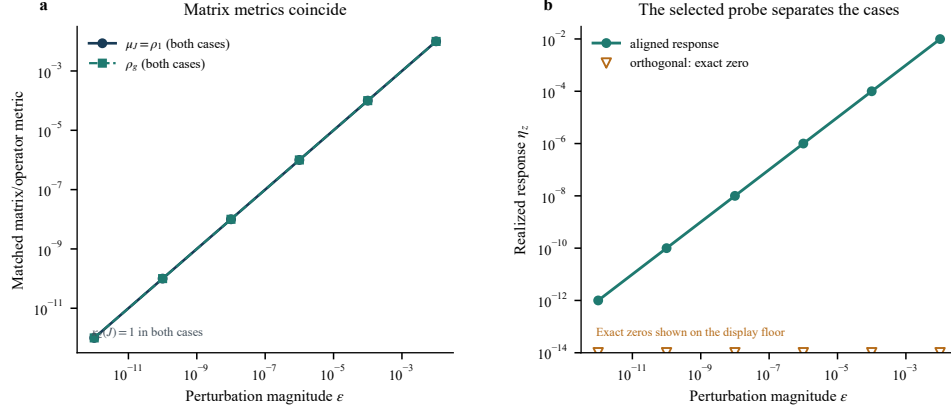
$$\begin{aligned}
 J &= I_2, & b &= e_1 = [1, 0]^T, \\
 \Delta J_A &= \varepsilon \operatorname{diag}(1, 0), & \Delta J_B &= \varepsilon \operatorname{diag}(0, 1), \quad \varepsilon > 0.
 \end{aligned}
 \tag{6}$$

183 The matrix and operator metrics are matched in the two cases. Since $\|J\| = 1$,
 184 both perturbations give $\mu_J = \varepsilon$; since $J^{-1} = I_2$, both give $\rho_1 = \varepsilon$; and since the
 185 guarded inverse scales only the perturbed coordinate, both give $\rho_g = \varepsilon/(1 + \varepsilon)$.
 186 The spectral condition number is one in each case. Thus even the combination
 187 of matrix amplitude, unguarded inverse amplification, guarded inverse amplifica-
 188 tion, and condition number is identical. This equality is stronger than matching
 189 only $\|\Delta J\|$: the two perturbations also have the same finite worst-direction bound.
 190 What remains unspecified by those metrics is whether the selected solution lies in
 191 the range of the perturbation. The construction therefore prepares a direct test of
 192 whether matched matrix-level and operator-level quantities determine one realized
 193 response.

$$\begin{aligned}
 \mu_J(A) &= \mu_J(B) = \varepsilon, \\
 \rho_1(A) &= \rho_1(B) = \varepsilon, \\
 \rho_g(A) &= \rho_g(B) = \frac{\varepsilon}{1 + \varepsilon}, \\
 \kappa_2(J) &= 1.
 \end{aligned}
 \tag{7}$$

195 The realized responses separate despite the matched metrics. In case A, the
 196 perturbation acts directly on $z = e_1$, and the guarded solution contracts the first
 197 component to $1/(1 + \varepsilon)$. The relative response is therefore $\varepsilon/(1 + \varepsilon)$, which at-
 198 tains the guarded-inverse bound. In case B, the perturbation acts only on the sec-
 199 ond coordinate, while the selected solution has a zero second component. Hence
 200 $\Delta J_B z = 0$, the guarded and unguarded solutions coincide, and $\eta_z(B) = 0$. P2 there-
 201 fore proves that matched matrix amplitude, condition number, and worst-direction
 202 amplification do not determine the realized response for a fixed probe. The missing
 203 information is directional alignment between the solution, the perturbation range,
 204 and the propagated response, not an additional scalar matrix magnitude.

Figure 3 contrasts the aligned and orthogonal constructions under matched matrix and worst-direction quantities.



Exact construction and registered reproduction; six frozen amplitudes; Euclidean/spectral norm

Figure 3: Aligned and orthogonal realized responses under matched matrix and worst-direction quantities for six pre-specified perturbation amplitudes.

$$\eta_z(A) = \frac{\varepsilon}{1 + \varepsilon}, \quad \eta_z(B) = 0. \quad (8)$$

The pre-specified numerical reproduction evaluated six frozen amplitudes from 1×10^{-12} to 1×10^{-2} using an independent high-precision route for the exact construction. Across all six rows, the maximum relative difference between the paired matrix metrics was 0, the maximum relative error in the nonzero aligned response was $8.721173782511926 \times 10^{-102}$, and the maximum absolute orthogonal response was 0. These values reproduce the analytical metric match and the response distinction without changing the construction after observation. The numerical verification validates implementation of this exact example; it does not add an empirical frequency statement. In particular, six amplitudes for one two-dimensional construction are not an operator-family survey, and the zero response in case B follows from exact orthogonality rather than from a general perturbation-screening rule.

$$\begin{aligned} \max \delta_{\text{matched}} &= 0, \\ \max \delta_{\eta_A} &= 8.721173782511926 \times 10^{-102}, \\ \max |\eta_B| &= 0. \end{aligned} \quad (9)$$

221 The logical content of P2 is an existence result. It shows that scalar matrix
 222 amplitude, a condition-number comparison, and even a matched worst-direction
 223 finite operator norm are insufficient to reconstruct η_z for one selected right-hand
 224 side. It does not estimate how often alignment or orthogonality occurs, assign
 225 probabilities to response levels, classify either perturbation for practical use, or
 226 infer behavior under spatial refinement. The pre-specified reproduction strength-
 227 ens confidence that the exact construction was evaluated as specified, but it does
 228 not enlarge the theorem's domain. Any claim about a family of coupled operators
 229 would require a separately designed family, declared sampling rule, and evidence
 230 appropriate to that population. Accordingly, P2 is used here only to establish non-
 231 determination of a directional response by the matched matrix metrics.

232 4. Coordinate-Transformation Verification

233 Let the physical unknown and scaled unknown be related by the nonsingular
 234 matrix D_x , and let residual coordinates be transformed by the nonsingular matrix
 235 D_r . The frozen convention is $z = D_x \hat{z}$ and $\hat{r} = D_r^{-1} r$. Substitution into $r = Jz - b$
 236 defines the scaled operator and right-hand side as $\hat{J} = D_r^{-1} J D_x$ and $\hat{b} = D_r^{-1} b$. A
 237 perturbation belongs to the same operator relation and must therefore transform
 238 as $\widehat{\Delta J} = D_r^{-1} \Delta J D_x$. These equations define equivalent coordinates only when the
 239 unknown, residual, right-hand side, operator, and perturbation are transformed
 240 together. Applying a transformation to some objects while independently redefin-
 241 ing another object creates a different perturbed problem, even if the same numeric
 242 symbol is displayed. Scaling and equilibration literature motivates explicit coor-
 243 dinate declarations but does not supply unique physical choices for D_x and D_r
 244 [8, 9, 19, 10].

$$\begin{aligned}
 z &= D_x \hat{z}, & \hat{r} &= D_r^{-1} r, \\
 \hat{J} &= D_r^{-1} J D_x, & \hat{b} &= D_r^{-1} b, \\
 \widehat{\Delta J} &= D_r^{-1} \Delta J D_x.
 \end{aligned}
 \tag{10}$$

246 Suppose first that the perturbation is declared in physical coordinates as
 247 $\Delta J_{\text{phys}} = \varepsilon I$. Its representation in the scaled operator is obtained by the joint
 248 transformation, giving $\widehat{\Delta J} = \varepsilon D_r^{-1} D_x$. The transformed perturbation is a scalar
 249 identity only when $D_r^{-1} D_x$ is a scalar multiple of the identity, and it retains the
 250 same εI representation only when $D_x = D_r$. In a mixed-dimensional system,
 251 different unknown and residual scales generally make the diagonal entries field

252 dependent. This is a coordinate statement: it identifies the scaled representation
 253 of one fixed physical operator. It neither changes that operator nor evaluates
 254 the practical consequences of its action. Treating the transformed matrix as if it
 255 had to retain the original scalar identity form would replace transformation by
 256 redeclaration.

$$257 \quad \Delta J_{\text{phys}} = \varepsilon I, \quad \widehat{\Delta J} = \varepsilon D_r^{-1} D_x. \quad (11)$$

258 The reverse construction begins with a perturbation declared in scaled coordi-
 259 nates as $\widehat{\Delta J} = \beta I$. Mapping it to physical coordinates gives $\Delta J_{\text{phys}} = \beta D_r D_x^{-1}$,
 260 which is generally a field-dependent diagonal operator. Thus a dimensionless iden-
 261 tity guard denotes a family of physical perturbations indexed by the declared un-
 262 known and residual scales. The mapping does not select those scales; it only states
 263 which physical operator corresponds to the scaled identity after the coordinate con-
 264 tract is fixed. Consequently, the label βI is incomplete without the operator space,
 265 D_x , D_r , and the direction in which the transformed solution is interpreted. This dis-
 266 tinction also prevents a same-numeric identity written in two coordinate systems
 267 from being treated as one perturbation without checking the transformation rule.

268 Figure 4 distinguishes equivalent coordinate transformation from same-
 269 numeric-label redeclaration.

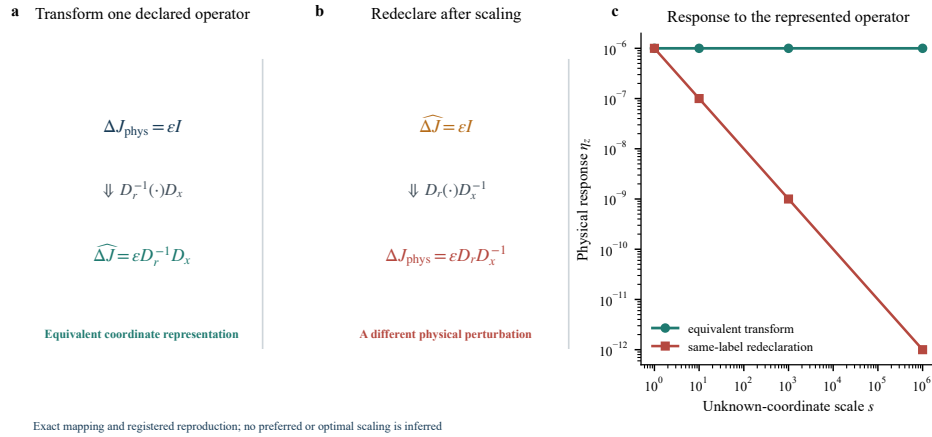


Figure 4: Equivalent coordinate transformation and same-numeric identity redeclaration under the declared coordinate convention.

$$270 \quad \widehat{\Delta J} = \beta I, \quad \Delta J_{\text{phys}} = \beta D_r D_x^{-1}. \quad (12)$$

271 The equivalent-transformation calculation used a two-dimensional construc-
 272 tion with pre-specified scale factors 1, 10, 1×10^3 , and 1×10^6 . In the equiva-
 273 lent transformed-operator route, the unknown, right-hand side, operator, and per-
 274 turbation were transformed consistently. The resulting maximum relative differ-
 275 ence between physical-equivalent perturbation operators was 7.490674676829096
 276 $\times 10^{-102}$, and the relative spread of the physical response was 0. These results vali-
 277 date the implemented transformation for the frozen construction and pre-specified
 278 scales. They do not establish that arbitrary independently specified guards are
 279 invariant; invariance here follows because the transformed descriptions represent
 280 the same physical operator and the same physical solution response. The numeri-
 281 cal values therefore check the algebraic mapping rather than define a coordinate-
 282 independent scalar guard.

$$\begin{aligned}
 \max \delta_{\text{operator}} &= 7.490674676829096 \times 10^{-102}, \\
 \text{spread}(\eta_Z)_{\text{transformed}} &= 0.
 \end{aligned}
 \tag{13}$$

284 The second pre-specified lane reapplied the same numeric identity after chang-
 285 ing coordinates instead of transforming the original perturbation. With $D_r = I_2$
 286 and $D_x = \text{diag}(1, s)$, that scaled identity maps back to a physical perturbation
 287 whose second diagonal entry is ε/s , whereas the original physical identity retains
 288 ε on both diagonal entries. The two physical perturbations coincide at $s = 1$ and
 289 differ for every pre-specified $s > 1$; the pre-specified inequality check was there-
 290 fore true. The same-numeric-label redeclaration result demonstrates that a shared
 291 numeric label does not establish operator equivalence. It does not show that one of
 292 the resulting operators should be selected. The observed response change follows
 293 from comparing different physical perturbations, not from a failure of coordinate
 294 transformation.

$$\Delta J_{\text{redeclared}} \neq \Delta J_{\text{transformed}} \quad \text{for every pre-specified } s > 1.
 \tag{14}$$

296 P3 establishes a declaration requirement, not a scale-selection method. Neither
 297 the algebraic mapping nor the pre-specified verification identifies unique phys-
 298 ical scales, an optimal equilibration, field weights, or a coordinate system with
 299 preferable solution behavior. Equilibration can alter numerical conditioning, while
 300 physical nondimensionalization can encode modeling choices, but those purposes
 301 require separate criteria and evidence [19, 10]. The result here is narrower: a per-
 302 turbation comparison is meaningful only after the coordinate transformation and

operator identity have been fixed. A future analysis may compare alternative scale declarations, but it must treat them as different study designs unless their operators, data, and response coordinates are transformed equivalently. This boundary prevents a coordinate label from being converted into an assessment of perturbation quality.

5. Uncertainty-Model Validation

Let z and z_g denote exact solutions in one declared truth lane, and let $\tilde{z}$ and $\tilde{z}_g$ be the corresponding computed parent solutions. Direct subtraction produces $\delta_{\text{obs}} = \text{fl}(\tilde{z}_g - \tilde{z})$, whereas the exact response is $\delta z = z_g - z$. The computed parents are represented as $\tilde{z} = z + e$ and $\tilde{z}_g = z_g + e_g$, and the rounded vector subtraction contributes an additional error q . These definitions separate the mathematical response from its computed observation. A direct difference can be interpreted only relative to bounds for both parent-solution errors and the subtraction error, all expressed in the same coordinates and norm. IEEE arithmetic and forward-error analysis motivate this separation, but they do not provide an application-specific budget without verified operands and solves [11, 12, 13].

$$\begin{aligned} \tilde{z} &= z + e, & \tilde{z}_g &= z_g + e_g, \\ \delta_{\text{obs}} &= \text{fl}(\tilde{z}_g - \tilde{z}) = \delta z + e_g - e + q. \end{aligned} \tag{15}$$

Proposition P4 assumes valid bounds $\|e\| \leq U_z$, $\|e_g\| \leq U_{z_g}$, and $\|q\| \leq U_{\text{sub}}$ and defines $U_{\text{direct}} = U_z + U_{z_g} + U_{\text{sub}}$. The triangle inequality then gives $\|\delta_{\text{obs}} - \delta z\| \leq U_{\text{direct}}$, while the forward and reverse triangle inequalities enclose the exact response norm between $\max(0, d_{\text{obs}} - U_{\text{direct}})$ and $d_{\text{obs}} + U_{\text{direct}}$, where $d_{\text{obs}} = \|\delta_{\text{obs}}\|$. If $d_{\text{obs}} > U_{\text{direct}}$, zero is excluded under that pre-specified error model. The conclusion is conditional: invalid or mismatched parent bounds invalidate the distinguishability statement even if the subtraction itself is evaluated exactly. A small linear residual or backward error alone is insufficient because forward error also depends on problem sensitivity [7, 13]. P4 therefore supplies no fixed binary64 threshold; it specifies what an uncertainty argument must cover.

$$\begin{aligned} U_{\text{direct}} &= U_z + U_{z_g} + U_{\text{sub}}, \\ \|\delta_{\text{obs}} - \delta z\| &\leq U_{\text{direct}}, \\ \max(0, d_{\text{obs}} - U_{\text{direct}}) &\leq \|\delta z\| \leq d_{\text{obs}} + U_{\text{direct}}. \end{aligned} \tag{16}$$

Definition (uncertainty-qualified distinguishability from zero).

Under a declared truth contract, response route, coordinate system, norm, and route-specific uncertainty model, a computed nonnegative response d_{obs} with certified absolute uncertainty bound U is *distinguishable from zero* when

$$d_{\text{obs}} - U > 0. \quad (17)$$

The statement excludes zero only under the declared uncertainty model. It does not establish a prescribed relative accuracy unless a separate accuracy criterion is declared, and it does not establish physical or engineering significance, guard safety or harm, or nonlinear consequence. The decision is route- and truth-contract-specific; it introduces no free numerical threshold.

The revised P4 uncertainty model implements this conditional logic through two non-interchangeable truth contracts. The decimal-source reference lane treats the pre-specified decimal operators and right-hand sides as the source problem, while the binary64-operand reference lane treats the exact reconstructed binary64 operands supplied to the solver as a separate real-valued problem. Writing $A_s x_s = b_s$ and $A_b x_b = b_b$ for the source and binary parent problems, respectively, and $\tilde{x}$ for either computed parent, each solution is compared with the residual and inverse norm of its own lane. The common subtraction allowance uses the frozen binary64 unit roundoff, a one-operation factor, the absolute parent operands, and an underflow allowance. The source and binary parent envelopes are then combined with the same subtraction term, but each total is compared only with its corresponding response truth. This non-mixing rule is necessary because decimal-to-binary representation and guarded-matrix construction can change the source problem before linear-solve error is assessed. Higher precision supports a reference only when its implementation and operation graph are independently checked [14, 15].

Figure 5 keeps the decimal-source and binary64-operand reference contracts separate.

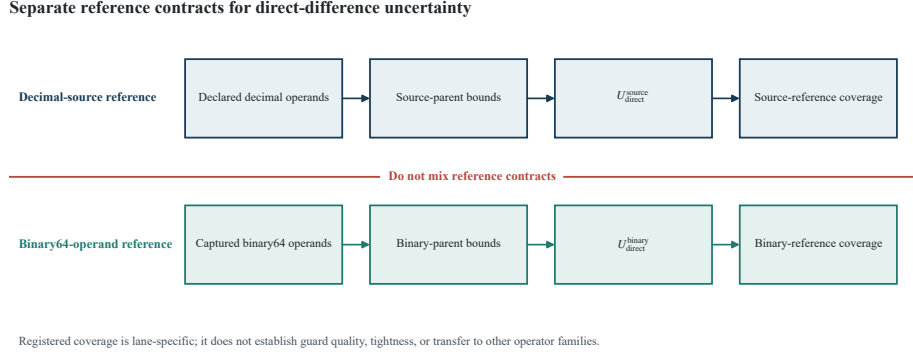


Figure 5: Separate decimal-source and binary64-operand reference contracts used to construct route-specific error enclosures.

$$\begin{aligned}
 A_s x_s &= b_s, & A_b x_b &= b_b, \quad u = 2^{-53}, \\
 \gamma_1 &= \frac{u}{1-u}, \\
 U_{\text{parent},s} &= \|A_s^{-1}\|_F \|b_s - A_s \tilde{x}\|_2, \\
 U_{\text{parent},b} &= \|A_b^{-1}\|_F \|b_b - A_b \tilde{x}\|_2, \\
 U_{\text{sub}} &= \gamma_1 \|\tilde{z}_g\| + \|\tilde{z}\|_2 + \sqrt{n} 2^{-1075}, \\
 U_s &= U_{\text{parent},s}(z) + U_{\text{parent},s}(z_g) + U_{\text{sub}}, \\
 U_b &= U_{\text{parent},b}(z) + U_{\text{parent},b}(z_g) + U_{\text{sub}}.
 \end{aligned} \tag{18}$$

Before revision, the pre-specified historical direct envelope was reproduced exactly on the 300-case development family. It covered 287/300 response differences and recorded three severe undercoverage cases. The frozen revision contract identified a mismatch in the historical truth specification: the historical parent-solve envelopes were formed relative to rounded binary operands, while the reference response was defined by the pre-specified decimal source operators. Under the decimal-source truth contract, the historical envelope therefore omitted a source-representation response component. This attribution motivated separate source and binary lanes; it did not change the historical result or assign the misses to the matrix perturbation itself. Exact reproduction of 287/300 and three severe cases was required before the revised model was evaluated.

The revised model was first evaluated on the frozen 300-case development

372 set comprising SPD, nonsymmetric, and saddle-point-like decimal families with
 373 dimensions 2, 4, 8, and 16. The source envelope covered 300/300 decimal-
 374 source reference-response errors, and the binary envelope covered 300/300
 375 binary64-operand reference-response errors. The pre-specified counts of severe
 376 undercoverage, parent-component bound failures, subtraction-bound failures,
 377 and nonfinite results were all zero. The maximum reference relative residual
 378 was $6.15236029869614753170882916397730867400586658 \times 10^{-105}$, below the
 379 pre-specified tolerance of 1×10^{-70} . These results validate source and binary
 380 error coverage for the development design. They do not evaluate bound tightness
 381 and do not extend the error model to operator dimensions outside the pre-specified
 382 family.

383 The disjoint holdout used the same three construction families and dimensions
 384 but a frozen independent seed, different condition-exponent grid, and different
 385 guard-amplitude grid. Its source envelope covered 300/300 decimal-source
 386 reference-response errors, and its binary envelope covered 300/300 binary64-
 387 operand reference-response errors. Severe undercoverage, component-bound
 388 failures, subtraction-bound failures, and nonfinite results again had zero counts,
 389 and two complete evaluator runs were exactly repeatable. The development
 390 and holdout results therefore provide complete pre-specified coverage on the
 391 development and holdout sets in both truth lanes. Because the holdout remains
 392 within the pre-specified construction families and dimensions, it tests disjoint
 393 pre-specified cases rather than unrestricted matrix systems. No extrapolation to
 394 dimensions 196 or 900 follows from these results.

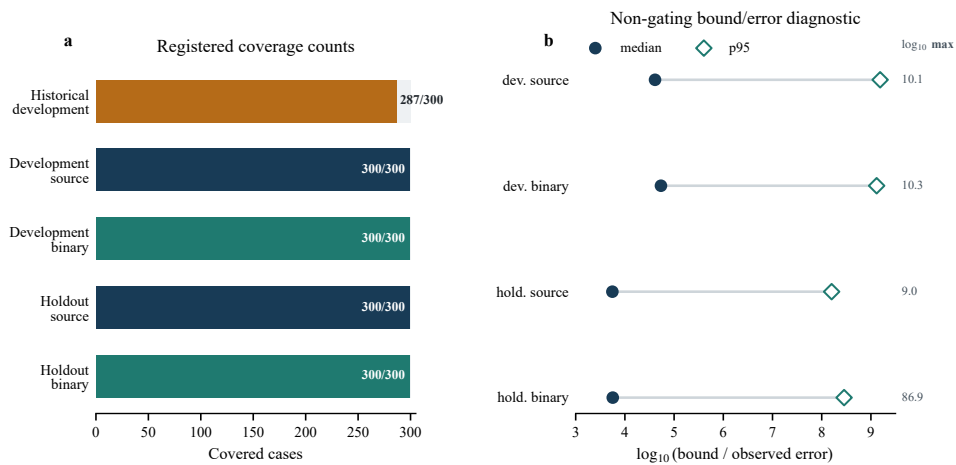
395 Table 2 summarizes historical reproduction and pre-specified development and
 396 holdout coverage.

397 Figure 6 reports development and holdout coverage separately from the de-
 398 scriptive tightness diagnostic.

399 Coverage and tightness answer different questions. The coverage criteria ask
 400 whether every modeled error is enclosed; they do not limit how conservative an
 401 enclosure may be. A post-execution descriptive diagnostic reported median bound-
 402 to-error ratios of 4.096571×10^4 and 5.405257×10^4 for the development source
 403 and binary lanes, respectively, and 5.565432×10^3 and 5.666318×10^3 for the
 404 holdout source and binary lanes. These ratios show that many bounds exceed their
 405 corresponding observed errors by substantial factors, as expected from a direction-
 406 independent inverse-Frobenius envelope applied to directional response errors. No
 407 tightness criterion was pre-specified, so the diagnostic cannot validate estimator

Table 2: Historical reproduction and development/holdout coverage in the two declared truth lanes.

Dataset/lane	Coverage	Severe undercoverage	Evidence role
historical development	287/300 (13 uncovered)	3	reproduced motivation
development source	300/300 (0 uncovered)	0	pre-specified coverage
development binary	300/300 (0 uncovered)	0	pre-specified coverage
holdout source	300/300 (0 uncovered)	0	pre-specified coverage
holdout binary	300/300 (0 uncovered)	0	pre-specified coverage



Coverage is registered; tightness remains unqualified.

Figure 6: Historical, development, and holdout coverage together with the non-gating bound-to-error diagnostic.

408 efficiency or class sharpness. The result supports decimal-source and binary64-
 409 operand reference-error coverage; the present limitation is that the absence of a
 410 tightness qualification remains in force.

411 Figure 7 records the source-contract attribution of the three historical severe-
 412 undercoverage cases without assigning a guard defect.



The two source-representation markers are registered components and are not summed. This explains the historical truth-contract mismatch; it does not attribute a guard defect.

Figure 7: Source-representation attribution for the three historical severe-undercoverage cases.

413 The exact P1 identity also permits a response solve, $(J + \Delta J)\delta z = -\Delta Jz$,
 414 that avoids direct subtraction of two full computed solutions. In a selected pre-
 415 specified small-perturbation subset, this route had a reported win rate of 1 relative
 416 to direct subtraction and a median error ratio of $6.835288346086798 \times 10^{-10}$. This
 417 observation is diagnostic motivation only. A computed response solve inherits er-
 418 ror from the parent solution used to form ΔJz , arithmetic used to construct that
 419 product, and the subsequent solve. It is therefore not an authoritative truth route
 420 without its own forward-error enclosure or independent higher-precision compar-
 421 ison [7, 13, 15]. Avoiding one subtraction changes the error pathway; it does not
 422 remove the need to validate that pathway.

423 The validated scope of the revised P4 uncertainty model is confined to the pre-
 424 specified SPD, nonsymmetric, and saddle-point-like decimal families, dimensions
 425 2, 4, 8, and 16, and their disjoint holdout, for a total of 600 evaluated cases. The
 426 construction supplies evidence across several operator forms, condition grids, and
 427 perturbation amplitudes, but it does not establish coverage for unrestricted oper-

428 ator populations or for larger frozen mixed-field tangents. In particular, no case-
 429 matched decimal-source reference verification was pre-specified for dimensions
 430 196 or 900, and no mesh-independent coverage law was tested. The framework
 431 can be applied to another operator only after its solved space, coordinates, norms,
 432 perturbation, right-hand side, parent error bounds, subtraction bound, precision,
 433 and truth route are declared and validated. This boundary keeps pre-specified cov-
 434 erage separate from future generalization. Coverage is jointly a property of the
 435 sampled operator family, the declared truth route, the precision used to construct
 436 the operands, and the particular error envelope; changing any of these objects de-
 437 fines a new validation problem rather than a continuation of the pre-specified one.
 438 The presence of SPD, nonsymmetric, and saddle-point-like families broadens the
 439 frozen test design, but finite samples from those families cannot establish behavior
 440 for every member of the corresponding mathematical classes. Likewise, disjoint
 441 holdout construction reduces reuse of the development grid but does not supply an
 442 external population model. Future extension to larger mixed-field systems would
 443 therefore require case-matched source and binary operands, an independent refer-
 444 ence route, pre-specified forward-error components, and a new holdout whose de-
 445 sign is fixed before response results are inspected. These requirements are method-
 446 ological consequences of the coverage claim’s scope, not evidence that the current
 447 envelopes will fail or succeed outside it.

448 The pre-specified validation scope was limited to decimal-source and binary64-
 449 operand reference-error coverage. All 300 development systems and all 300 hold-
 450 out systems were covered in each truth lane, with no severe undercoverage. The
 451 study did not qualify the tightness of the resulting envelopes.

452 **6. Applications to Two Coupled Finite-Element Linearizations**

453 *6.1. Frozen operators and route definitions*

454 The application stage uses two independently frozen transformed free-system
 455 snapshots. M8 is a 196×196 nonsymmetric tangent with 5442 stored nonzeros;
 456 M16 is a 900×900 nonsymmetric tangent with 28259 stored nonzeros. Each
 457 snapshot fixes the IEEE binary64 matrices J and J_g , the common right-hand side
 458 b , the free-degree-of-freedom ordering, the unknown and residual scales, and the
 459 Euclidean comparison norm before either response route is evaluated. The pertur-
 460 bation is reconstructed as $\Delta J = J_g - J$ and is diagonal in both snapshots. Figure 8

461 shows the two sparsity patterns and their independently qualified response inter-
 462 vals.

463 The truth object for each application is the real-valued system represented ex-
 464 actly by the captured binary64 operands. Integer-ratio reconstruction, signed-zero
 465 checks, and the identity $J_g = J + \Delta J$ establish that operand contract before high-
 466 precision evaluation. Parent systems and the response equation are then solved
 467 independently at 120 and 180 decimal digits. The direct-difference route evalu-
 468 ates $\text{fl}(\tilde{z}_g - \tilde{z})$ and includes both parent-solution and subtraction components. The
 469 response-equation route evaluates $(J + \Delta J)\delta z = -\Delta Jz$ and uses its own parent,
 470 forcing-construction, and response-solve components. Each route is compared
 471 only with its matching uncertainty enclosure.

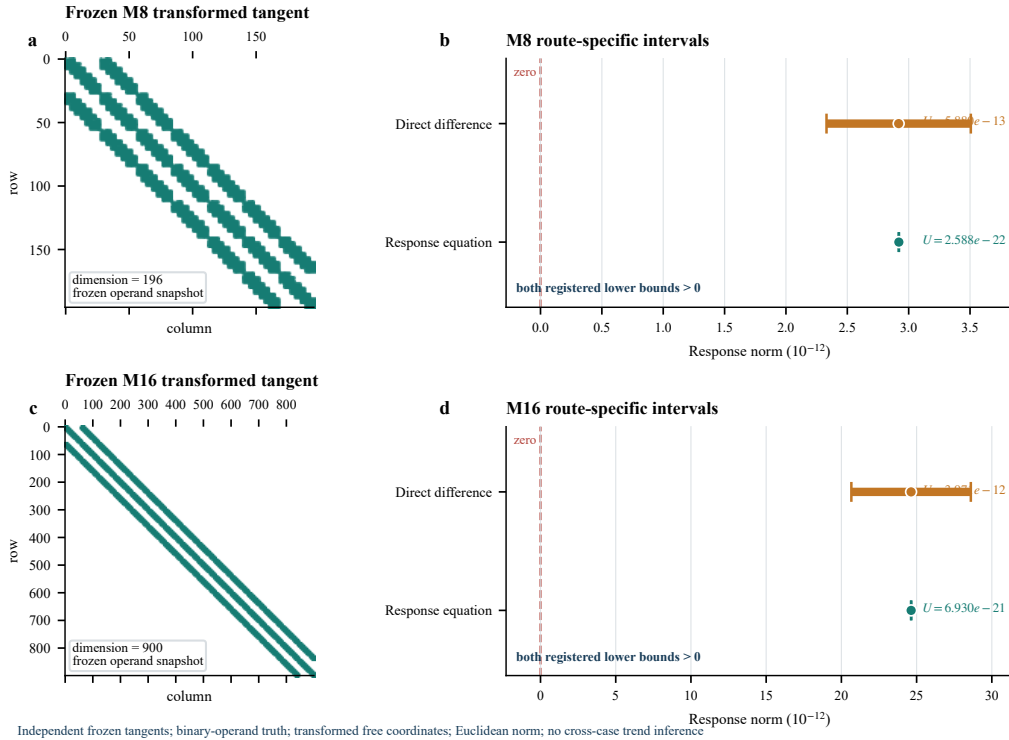


Figure 8: Route-specific response qualification for two independently frozen coupled finite-element tangents. Panels show the M8 and M16 sparsity patterns and the direct-difference and response-equation intervals; dashed references denote zero.

472 6.2. M8 application

473 For M8, the maximum normalized difference between the 120- and 180-digit
474 references was $8.805709238088443 \times 10^{-121}$, the maximum reference relative
475 residual was $2.550975633144060 \times 10^{-124}$, and the response-identity difference
476 was $9.184908547614584 \times 10^{-123}$. Independent-process repetitions reproduced
477 the reference and binary-route artifacts exactly.

478 The direct-difference route produced $d_{\text{obs}} = 2.918501453675816 \times 10^{-12}$
479 and $U = 5.880137782404462 \times 10^{-13}$, giving the positive lower bound
480 $2.330487675435369 \times 10^{-12}$ and a relative-error upper bound of approximately
481 0.2523136. The response-equation route produced $d_{\text{obs}} = 2.920014225448921 \times$
482 10^{-12} and $U = 2.588299567178917 \times 10^{-22}$, again giving a positive lower bound.
483 Both M8 responses are therefore distinguishable from zero under their respective
484 route-specific enclosures.

485 6.3. M16 application

486 For M16, the maximum 120/180-digit normalized difference was

$$3.970070132750439 \times 10^{-121}.$$

487 The maximum reference relative residual was $4.144081636789256 \times 10^{-124}$, and
488 the response-identity difference was $3.042263371127549 \times 10^{-122}$. A separate
489 process reproduced both precision references, route artifacts, and decisions exactly,
490 with no mutation of the operand snapshot.

491 The direct-difference route produced $d_{\text{obs}} = 2.463376440770722 \times 10^{-11}$ and
492 $U = 3.970799910942942 \times 10^{-12}$, giving the lower bound 2.066296449676427
493 $\times 10^{-11}$ and a relative-error upper bound of approximately 0.1921699. The
494 response-equation route produced $d_{\text{obs}} = 2.463671846097062 \times 10^{-11}$ and $U =$
495 $6.930482617167356 \times 10^{-21}$, giving the positive lower bound 2.463671845404014
496 $\times 10^{-11}$. Both M16 responses are distinguishable from zero under their own en-
497 closures. Exact decimal strings and uncertainty components are retained in
498 Supplementary Tables S4–S6.

499 6.4. Replication and offline cost

500 Table 3 summarizes the four case-route results without combining them into
501 an estimator score. Completing the same declaration, reference, uncertainty, and

zero-exclusion sequence for both M8 and M16 demonstrates replication of the assessment procedure on two frozen coupled finite-element linearizations. The cases were not selected or controlled as a mesh, dimension, or conditioning sequence, so their values do not define a cross-case trend or a universal ranking of the arithmetic routes.

Table 3: Route-specific response qualification for the two frozen finite-element tangents. Displayed values are rounded; Supplementary Table S4 retains the exact decimal strings.

Case	Dimension	Route	Observed response	Uncertainty bound	Lower bound	Rel.-error upper bound	Decision
M8	196	Direct difference	2.918501×10^{-12}	5.880138×10^{-13}	2.330488×10^{-12}	2.523136×10^{-1}	Distinguishable
M8	196	Response equation	2.920014×10^{-12}	2.588300×10^{-22}	2.920014×10^{-12}	8.863996×10^{-11}	Distinguishable
M16	900	Direct difference	2.463376×10^{-11}	3.970800×10^{-12}	2.066296×10^{-11}	1.921699×10^{-1}	Distinguishable
M16	900	Response equation	2.463672×10^{-11}	6.930483×10^{-21}	2.463672×10^{-11}	2.813071×10^{-10}	Distinguishable

The complete M8 offline calculation, including both precision levels and an independent repeat, required 467.5920785 s and 115154944 bytes of peak memory in the recorded environment. M16 required 37512.1609418 s and 1452195840 bytes. These two observations are computational provenance rather than a complexity fit. They show that the present high-precision route is an offline qualification oracle. Sparse updates, inverse-norm estimators, verified sparse solvers, and precision-aware refinement offer relevant directions for a scalable estimator [20, 21, 16, 22], but any replacement route would require an independent accuracy and coverage study.

7. Discussion

7.1. Why matrix magnitude is insufficient

The analysis separates questions that are often compressed into one statement about a small matrix term. The ratio μ_J measures amplitude in declared coordinates, ρ_1 and ρ_g measure inverse-mediated worst-direction amplification, and η_z measures the response of one selected right-hand side. Proposition P2 shows that even matched values of μ_J , ρ_1 , ρ_g , and $\kappa(J)$ do not determine η_z . Directional information enters through ΔJz and its propagation by the perturbed inverse. Condition-number comparisons therefore remain useful bounds, but not directional mechanism classifiers [5, 6, 7].

7.2. Why coordinate declaration matters

The matrix representation of a perturbation changes with unknown and residual coordinates. Under $z = D_x \hat{z}$ and $\hat{r} = D_r^{-1} r$, a physical identity perturbation

529 becomes $\varepsilon D_r^{-1} D_x$, while a dimensionless identity maps back as $\beta D_r D_x^{-1}$. The co-
530 ordinate verification confirms equivalent responses only when the operator, data,
531 and perturbation are transformed together. Reusing the same numeric identity after
532 changing coordinates declares a different physical operator. The procedure con-
533 sequently requires a solved-space and coordinate contract before any matrix ratio
534 or response comparison is interpreted; it does not prescribe a preferred nondimen-
535 sionalization or equilibration [8, 9, 19, 10, 3].

536 7.3. *Why distinguishability differs from accuracy and physics*

537 The test $d_{\text{obs}} - U > 0$ answers one numerical question: whether zero is ex-
538 cluded under the declared truth object, arithmetic route, and uncertainty model.
539 Coverage asks whether the model enclosed known reference errors over a specified
540 family, whereas relative accuracy requires an additional ratio criterion. Physical
541 significance requires a problem-dependent response scale or consequence. These
542 are separate statements. The complete development and holdout coverage sup-
543 ports the two declared truth lanes within the evaluated families; the finite-element
544 applications support route-specific zero exclusion for their captured operands.

545 7.4. *Why arithmetic routes require separate uncertainty models*

546 Direct subtraction and the response equation implement the same exact re-
547 sponse identity through different floating-point operation graphs. Direct subtrac-
548 tion inherits the errors of two parent solves and the subtraction itself. The response
549 equation inherits error from the parent used to construct ΔJz , the forcing construc-
550 tion, and the response solve. A tighter interval for one route in M8 or M16 is
551 therefore a case-specific consequence of these decompositions. It cannot establish
552 a universal route ordering. This separation is also why small parent residuals alone
553 do not qualify a computed difference: a backward-error statement does not supply
554 every forward-error component required by the response route [7, 11, 12, 13].

555 7.5. *Replication across two finite-element linearizations*

556 The same ten-stage procedure was completed independently for the 196- and
557 900-dimensional tangent snapshots. Both applications fixed the operand and coor-
558 dinate contracts before route evaluation, reconstructed binary64 operands exactly,
559 used independent precision ladders, and qualified both response routes through
560 their own enclosures. This closes the methodological question of whether the com-
561 plete procedure can be repeated on more than one coupled finite-element lineariza-
562 tion. Because the two snapshots do not form a designed sequence or statistical

563 sample, the replication does not estimate a mesh law, dimension dependence, or
564 frequency over an operator population.

565 7.6. *Offline oracle versus scalable estimator*

566 Independent dense high-precision solves provide a clear offline truth route but
567 do not scale as an online production diagnostic. The recorded M8 and M16 costs
568 make this limitation visible without defining a performance law. Scalable alterna-
569 tives may combine sparse factor updates, inverse-norm estimation, verified sparse
570 solves, or precision-aware refinement [14, 20, 21, 16, 22]. Translating the present
571 method to such an estimator requires more than replacing the arithmetic kernel:
572 the estimator must preserve the solved-space and coordinate declarations, expose
573 route-specific uncertainty, and be validated for coverage against an independent
574 reference.

575 7.7. *Limitations*

576 The method assesses a tangent-only matrix perturbation at a declared lineariza-
577 tion. It does not convert that perturbation into a residual-consistent stabilization
578 and does not infer mixed-method stability, conservation, boundedness, physical
579 importance, or a subsequent nonlinear trajectory. Synthetic uncertainty coverage
580 is confined to the pre-specified operator families and dimensions, while the appli-
581 cation evidence is confined to two frozen binary64 snapshots. Table 4 summarizes
582 these interpretation boundaries once, rather than attaching the full list to every re-
583 sult.

584 Within these limits, the contribution is a computational assessment procedure
585 rather than a perturbation-selection algorithm. It makes the represented operator
586 explicit, separates matrix, operator, and directional quantities, and requires uncer-
587 tainty matched to the arithmetic route before a computed response is classified.

Table 4: Interpretation boundary by claim domain. Supplementary Table S7 retains the complete claim–evidence matrix.

Domain	Established	Not established
Theory	Finite response identity, directional non-determination, and coordinate mapping under declared assumptions.	Prevalence, optimal scaling, or a universal guard rule.
Synthetic evidence	Pre-specified reproduction and coverage on the frozen development and disjoint holdout designs.	Unrestricted-system coverage or estimator tightness.
Frozen tangents	Route-specific qualification completed separately for M8 and M16.	Mesh law, dimension trend, or population-level generality.
Route comparison	Each route has its own uncertainty components and a case-specific decision.	Universal route ranking or superiority.
Interpretation	Numerical distinguishability under declared contracts.	Guard safety or harm, nonlinear consequence, finite-element stability, or engineering qualification.
Cost	Recorded offline cost and resource provenance for each frozen case.	Online estimator feasibility, complexity fit, or production scaling.

588 8. Conclusions

589 This study developed a scale-explicit and uncertainty-qualified procedure for
590 assessing a tangent-only matrix perturbation in a coupled finite-element lineariza-
591 tion. The procedure declares the solved space and coordinates, separates matrix
592 amplitude from inverse-mediated and directional response measures, and qualifies
593 the selected response through a route-specific error enclosure.

594 An exact finite identity establishes the response relation, while a matched two-
595 dimensional construction proves that matrix and worst-direction quantities do not
596 determine one realized response. Coordinate-transformation verification shows
597 why identity perturbations require explicit operator-space declarations. Devel-
598 opment and disjoint holdout families then validate the two-contract uncertainty
599 model within their specified operator classes and dimensions.

600 The complete procedure was independently applied to frozen 196- and
601 900-dimensional coupled finite-element tangents. Direct-difference and response-
602 equation intervals excluded zero in both applications under their own uncertainty
603 models. The two cases establish replication of the method, while their route
604 values remain application specific.

605 The present high-precision implementation is an offline qualification oracle,
606 and the two frozen applications do not establish population-level generality or
607 nonlinear and engineering consequences. Future work should address scalable
608 route-specific uncertainty estimators for large sparse nonlinear finite-element lin-
609 earizations rather than extending the current offline oracle by brute force.

610 Data availability

611 The non-confidential frozen operands, synthetic validation records, exact-
612 value tables, claim–evidence matrix, and provenance files supporting this study
613 are archived in Zenodo at <https://doi.org/10.5281/zenodo.21833851> [23]. The
614 archive excludes Abaqus executables, user-element source, solver state histories,
615 and engineering project data.

616 Code availability

617 The scripts for the analytical constructions, uncertainty-model validation,
618 and the two offline finite-element linearization applications are available at
619 <https://github.com/yqma7980/scale-explicit-matrix-perturbation-observability>

620 and are preserved with the cited Zenodo release. Code is licensed under the MIT
621 License; data and documentation are licensed under CC BY 4.0.

622 **Reproducibility statement**

623 The reported calculations use immutable operand snapshots, explicit coordi-
624 nate and truth contracts, two independent precision levels, independent-process
625 repeatability checks, and exact output records. The Supplementary Information re-
626 tains full numerical precision, uncertainty components, and execution provenance.

627 **CRedit authorship contribution statement**

628 **Bing Liang:** Conceptualization, Methodology, Formal analysis, Writing –
629 original draft, Writing – review and editing, Supervision. **Yangqi Ma:** Methodol-
630 ogy, Software, Validation, Formal analysis, Investigation, Data curation, Visual-
631 ization, Writing – original draft, Writing – review and editing. **Weiji Sun:** Concep-
632 tualization, Methodology, Resources, Supervision, Project administration, Fund-
633 ing acquisition, Writing – review and editing. **Shi He:** Methodology, Validation,
634 Writing – review and editing.

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639 The authors declare that they have no known competing financial interests or
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642 **Declaration of generative AI and AI-assisted technologies in the writing pro-** 643 **cess**

644 During preparation of this work, the authors used Codex (OpenAI) to assist
645 with language revision, document organization, and consistency checks. The au-
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